



MATH240: Higher Order Odes

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Name: _____

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Characteristic equation

1. Find the roots of the characteristic equation for $y'' + 4y' - 5y = 0$.

$$x^2 + 4x - 5 = 0$$

Answer: _____

2. The motion of a spring-mass system satisfies $y'' + 12y' + 32y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 12x + 32 = 0$$

Answer: _____

3. Find the characteristic roots of $y'' + 4y' + 3y = 0$. Write the general solution.

$$x^2 + 4x + 3 = 0$$

Answer: _____

4. Solve the characteristic equation: $r^2 + 2r + 0 = 0$.

$$x^2 + 2x = 0$$

Answer: _____

5. Find the roots of the characteristic equation for $y'' + 1y' - 20y = 0$.

$$x^2 + 1x - 20 = 0$$

Answer: _____



6. The motion of a spring-mass system satisfies $y'' + 10y' + 21y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 10x + 21 = 0$$

Answer: _____

7. Find the characteristic roots of $y'' + 8y' + 12y = 0$. Write the general solution.

$$x^2 + 8x + 12 = 0$$

Answer: _____

8. Solve the characteristic equation: $r^2 + 6r + 5 = 0$.

$$x^2 + 6x + 5 = 0$$

Answer: _____

9. Find the roots of the characteristic equation for $y'' + -1y' + -6y = 0$.

$$x^2 - 1x - 6 = 0$$

Answer: _____

10. The motion of a spring-mass system satisfies $y'' + 11y' + 30y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 11x + 30 = 0$$

Answer: _____

11. Find the characteristic roots of $y'' + 10y' + 21y = 0$. Write the general solution.

$$x^2 + 10x + 21 = 0$$

Answer: _____

12. Solve the characteristic equation: $r^2 + 0r + -9 = 0$.

$$x^2 - 9 = 0$$

Answer: _____

13. Find the roots of the characteristic equation for $y'' + 0y' + -16y = 0$.

$$x^2 - 16 = 0$$

Answer: _____



14. The motion of a spring-mass system satisfies $y'' + 12y' + 35y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 12x + 35 = 0$$

Answer: _____

15. Find the characteristic roots of $y'' + 11y' + 28y = 0$. Write the general solution.

$$x^2 + 11x + 28 = 0$$

Answer: _____

16. Solve the characteristic equation: $r^2 + 7r + 6 = 0$.

$$x^2 + 7x + 6 = 0$$

Answer: _____

17. Find the roots of the characteristic equation for $y'' + -1y' + -2y = 0$.

$$x^2 - 1x - 2 = 0$$

Answer: _____

18. The motion of a spring-mass system satisfies $y'' + 12y' + 35y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 12x + 35 = 0$$

Answer: _____

19. Find the characteristic roots of $y'' + 8y' + 15y = 0$. Write the general solution.

$$x^2 + 8x + 15 = 0$$

Answer: _____

20. Solve the characteristic equation: $r^2 + 3r + -4 = 0$.

$$x^2 + 3x - 4 = 0$$

Answer: _____

21. Find the roots of the characteristic equation for $y'' + -1y' + -2y = 0$.

$$x^2 - 1x - 2 = 0$$

Answer: _____



22. The motion of a spring-mass system satisfies $y'' + 2y' + 1y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 2x + 1 = 0$$

Answer: _____

23. Find the characteristic roots of $y'' + 7y' + 0y = 0$. Write the general solution.

$$x^2 + 7x = 0$$

Answer: _____

24. Solve the characteristic equation: $r^2 + 2r + 0 = 0$.

$$x^2 + 2x = 0$$

Answer: _____

25. Find the roots of the characteristic equation for $y'' + 2y' + -15y = 0$.

$$x^2 + 2x - 15 = 0$$

Answer: _____

26. The motion of a spring-mass system satisfies $y'' + 14y' + 49y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 14x + 49 = 0$$

Answer: _____

27. Find the characteristic roots of $y'' + 2y' + 0y = 0$. Write the general solution.

$$x^2 + 2x = 0$$

Answer: _____

28. Solve the characteristic equation: $r^2 + 2r + -24 = 0$.

$$x^2 + 2x - 24 = 0$$

Answer: _____

29. Find the roots of the characteristic equation for $y'' + 2y' + -15y = 0$.

$$x^2 + 2x - 15 = 0$$

Answer: _____



30. The motion of a spring-mass system satisfies $y'' + 11y' + 24y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 11x + 24 = 0$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Characteristic equation. All answers verified by independent computation.

Solutions



Characteristic equation

1. Find the roots of the characteristic equation for $y'' + 4y' + -5y = 0$.

$$x^2 + 4x - 5 = 0$$

→ Characteristic equation: $r^2 + 4r + -5 = 0$.

→ Factor or use quadratic formula.

→ Roots: $r = -5, r = 1$.

→ General solution: $y = C_1 \cdot e^{(-5t)} + C_2 \cdot e^{(1t)}$.

Answer: $x = -5$ or $x = 1$

2. The motion of a spring-mass system satisfies $y'' + 12y' + 32y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 12x + 32 = 0$$

→ Characteristic equation: $r^2 + 12r + 32 = 0$.

→ Roots: $r_1 = -8, r_2 = -4$.

→ Both roots real and negative: overdamped system.

Answer: $x = -8$ or $x = -4$

3. Find the characteristic roots of $y'' + 4y' + 3y = 0$. Write the general solution.

$$x^2 + 4x + 3 = 0$$

→ $r^2 + 4r + 3 = 0$.

→ $r_1 = -3, r_2 = -1$.

→ General solution: $y = C_1 \cdot e^{(-3t)} + C_2 \cdot e^{(-1t)}$.

Answer: $x = -3$ or $x = -1$

4. Solve the characteristic equation: $r^2 + 2r + 0 = 0$.

$$x^2 + 2x = 0$$

→ Factor: $(r - -2)(r - 0) = 0$.

→ Roots: $r = -2$ and $r = 0$.

Answer: $x = -2$ or $x = 0$

5. Find the roots of the characteristic equation for $y'' + 1y' + -20y = 0$.

$$x^2 + 1x - 20 = 0$$

→ Characteristic equation: $r^2 + 1r + -20 = 0$.

→ Factor or use quadratic formula.

→ Roots: $r = -5, r = 4$.

→ General solution: $y = C_1 \cdot e^{(-5t)} + C_2 \cdot e^{(4t)}$.

Answer: $x = -5$ or $x = 4$



6. The motion of a spring-mass system satisfies $y'' + 10y' + 21y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 10x + 21 = 0$$

→ Characteristic equation: $r^2 + 10r + 21 = 0$.

→ Roots: $r_1 = -7$, $r_2 = -3$.

→ Both roots real and negative: overdamped system.

Answer: $x = -7$ or $x = -3$

7. Find the characteristic roots of $y'' + 8y' + 12y = 0$. Write the general solution.

$$x^2 + 8x + 12 = 0$$

→ $r^2 + 8r + 12 = 0$.

→ $r_1 = -6$, $r_2 = -2$.

→ General solution: $y = C_1 e^{-6t} + C_2 e^{-2t}$.

Answer: $x = -6$ or $x = -2$

8. Solve the characteristic equation: $r^2 + 6r + 5 = 0$.

$$x^2 + 6x + 5 = 0$$

→ Factor: $(r - -5)(r - -1) = 0$.

→ Roots: $r = -5$ and $r = -1$.

Answer: $x = -5$ or $x = -1$

9. Find the roots of the characteristic equation for $y'' + -1y' + -6y = 0$.

$$x^2 - 1x - 6 = 0$$

→ Characteristic equation: $r^2 + -1r + -6 = 0$.

→ Factor or use quadratic formula.

→ Roots: $r = -2$, $r = 3$.

→ General solution: $y = C_1 \cdot e^{-2t} + C_2 \cdot e^{3t}$.

Answer: $x = -2$ or $x = 3$

10. The motion of a spring-mass system satisfies $y'' + 11y' + 30y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 11x + 30 = 0$$

→ Characteristic equation: $r^2 + 11r + 30 = 0$.

→ Roots: $r_1 = -6$, $r_2 = -5$.

→ Both roots real and negative: overdamped system.

Answer: $x = -6$ or $x = -5$

11. Find the characteristic roots of $y'' + 10y' + 21y = 0$. Write the general solution.

$$x^2 + 10x + 21 = 0$$

→ $r^2 + 10r + 21 = 0$.

→ $r_1 = -7$, $r_2 = -3$.

→ General solution: $y = C_1 e^{-7t} + C_2 e^{-3t}$.

Answer: $x = -7$ or $x = -3$



12. Solve the characteristic equation: $r^2 + 0r + -9 = 0$.

$$x^2 - 9 = 0$$

→ Factor: $(r - -3)(r - 3) = 0$.

→ Roots: $r = -3$ and $r = 3$.

Answer: $x = -3$ or $x = 3$

13. Find the roots of the characteristic equation for $y'' + 0y' + -16y = 0$.

$$x^2 - 16 = 0$$

→ Characteristic equation: $r^2 + 0r + -16 = 0$.

→ Factor or use quadratic formula.

→ Roots: $r = -4$, $r = 4$.

→ General solution: $y = C1 \cdot e^{(-4t)} + C2 \cdot e^{(4t)}$.

Answer: $x = -4$ or $x = 4$

14. The motion of a spring-mass system satisfies $y'' + 12y' + 35y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 12x + 35 = 0$$

→ Characteristic equation: $r^2 + 12r + 35 = 0$.

→ Roots: $r1 = -7$, $r2 = -5$.

→ Both roots real and negative: overdamped system.

Answer: $x = -7$ or $x = -5$

15. Find the characteristic roots of $y'' + 11y' + 28y = 0$. Write the general solution.

$$x^2 + 11x + 28 = 0$$

→ $r^2 + 11r + 28 = 0$.

→ $r1 = -7$, $r2 = -4$.

→ General solution: $y = C1 \cdot e^{(-7t)} + C2 \cdot e^{(-4t)}$.

Answer: $x = -7$ or $x = -4$

16. Solve the characteristic equation: $r^2 + 7r + 6 = 0$.

$$x^2 + 7x + 6 = 0$$

→ Factor: $(r - -6)(r - -1) = 0$.

→ Roots: $r = -6$ and $r = -1$.

Answer: $x = -6$ or $x = -1$

17. Find the roots of the characteristic equation for $y'' + -1y' + -2y = 0$.

$$x^2 - 1x - 2 = 0$$

→ Characteristic equation: $r^2 + -1r + -2 = 0$.

→ Factor or use quadratic formula.

→ Roots: $r = -1$, $r = 2$.

→ General solution: $y = C1 \cdot e^{(-1t)} + C2 \cdot e^{(2t)}$.

Answer: $x = -1$ or $x = 2$



18. The motion of a spring-mass system satisfies $y'' + 12y' + 35y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 12x + 35 = 0$$

→ Characteristic equation: $r^2 + 12r + 35 = 0$.

→ Roots: $r_1 = -7$, $r_2 = -5$.

→ Both roots real and negative: overdamped system.

Answer: $x = -7$ or $x = -5$

19. Find the characteristic roots of $y'' + 8y' + 15y = 0$. Write the general solution.

$$x^2 + 8x + 15 = 0$$

→ $r^2 + 8r + 15 = 0$.

→ $r_1 = -5$, $r_2 = -3$.

→ General solution: $y = C_1 e^{(-5)t} + C_2 e^{(-3)t}$.

Answer: $x = -5$ or $x = -3$

20. Solve the characteristic equation: $r^2 + 3r - 4 = 0$.

$$x^2 + 3x - 4 = 0$$

→ Factor: $(r - -4)(r - 1) = 0$.

→ Roots: $r = -4$ and $r = 1$.

Answer: $x = -4$ or $x = 1$

21. Find the roots of the characteristic equation for $y'' + -1y' + -2y = 0$.

$$x^2 - 1x - 2 = 0$$

→ Characteristic equation: $r^2 + -1r + -2 = 0$.

→ Factor or use quadratic formula.

→ Roots: $r = -1$, $r = 2$.

→ General solution: $y = C_1 \cdot e^{(-1)t} + C_2 \cdot e^{(2)t}$.

Answer: $x = -1$ or $x = 2$

22. The motion of a spring-mass system satisfies $y'' + 2y' + 1y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 2x + 1 = 0$$

→ Characteristic equation: $r^2 + 2r + 1 = 0$.

→ Roots: $r_1 = -1$, $r_2 = -1$.

→ Both roots real and negative: overdamped system.

Answer: $x = -1$ or $x = -1$

23. Find the characteristic roots of $y'' + 7y' + 0y = 0$. Write the general solution.

$$x^2 + 7x = 0$$

→ $r^2 + 7r + 0 = 0$.

→ $r_1 = -7$, $r_2 = 0$.

→ General solution: $y = C_1 e^{(-7)t} + C_2 e^{(0)t}$.

Answer: $x = -7$ or $x = 0$



24. Solve the characteristic equation: $r^2 + 2r + 0 = 0$.

$$x^2 + 2x = 0$$

→ Factor: $(r - -2)(r - 0) = 0$.

→ Roots: $r = -2$ and $r = 0$.

Answer: $x = -2$ or $x = 0$

25. Find the roots of the characteristic equation for $y'' + 2y' + -15y = 0$.

$$x^2 + 2x - 15 = 0$$

→ Characteristic equation: $r^2 + 2r + -15 = 0$.

→ Factor or use quadratic formula.

→ Roots: $r = -5$, $r = 3$.

→ General solution: $y = C1 \cdot e^{(-5t)} + C2 \cdot e^{(3t)}$.

Answer: $x = -5$ or $x = 3$

26. The motion of a spring-mass system satisfies $y'' + 14y' + 49y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 14x + 49 = 0$$

→ Characteristic equation: $r^2 + 14r + 49 = 0$.

→ Roots: $r1 = -7$, $r2 = -7$.

→ Both roots real and negative: overdamped system.

Answer: $x = -7$ or $x = -7$

27. Find the characteristic roots of $y'' + 2y' + 0y = 0$. Write the general solution.

$$x^2 + 2x = 0$$

→ $r^2 + 2r + 0 = 0$.

→ $r1 = -2$, $r2 = 0$.

→ General solution: $y = C1 \cdot e^{(-2t)} + C2 \cdot e^{(0t)}$.

Answer: $x = -2$ or $x = 0$

28. Solve the characteristic equation: $r^2 + 2r + -24 = 0$.

$$x^2 + 2x - 24 = 0$$

→ Factor: $(r - -6)(r - 4) = 0$.

→ Roots: $r = -6$ and $r = 4$.

Answer: $x = -6$ or $x = 4$

29. Find the roots of the characteristic equation for $y'' + 2y' + -15y = 0$.

$$x^2 + 2x - 15 = 0$$

→ Characteristic equation: $r^2 + 2r + -15 = 0$.

→ Factor or use quadratic formula.

→ Roots: $r = -5$, $r = 3$.

→ General solution: $y = C1 \cdot e^{(-5t)} + C2 \cdot e^{(3t)}$.

Answer: $x = -5$ or $x = 3$



30. The motion of a spring-mass system satisfies $y'' + 11y' + 24y = 0$. Find the characteristic roots and identify the type of damping.

$$x^2 + 11x + 24 = 0$$

→ Characteristic equation: $r^2 + 11r + 24 = 0$.

→ Roots: $r_1 = -8$, $r_2 = -3$.

→ Both roots real and negative: overdamped system.

Answer: $x = -8$ or $x = -3$

